

Milestone 2: Inventory Intelligence & Customer Analytics

This milestone extends the marketplace by adding real customer transactions and turning transaction data into useful analytics and AI-driven insights.

- Extended ShopSense with a customer-facing marketplace and real order flow.
- Connected customer purchases with transaction and inventory data.
- Added customer segmentation and sales-based product recommendations.
- Added AI features for demand forecasting and review sentiment analysis.

Customer Portal & Transactions

- Customer enters basic details to start shopping.
- Browses the marketplace and places an order.
- The system stores the transaction automatically.
- Product stock is instantly reduced in the PostgreSQL database.

Insert Customer Portal / Marketplace Screenshot Here

Customer Analytics & Segmentation

- Customer spending is calculated directly from actual transaction records.
- Customers are automatically grouped into segments based on total spending:
 - High Value: \$500+
 - Regular: \$100-\$499
 - Low Value: below \$100
- This helps vendors understand who their top buyers are.

Insert Customer Analytics Screenshot Here

Gemini Review Sentiment Analysis

- The vendor selects a product and enters a customer rating and review text.
- Gemini AI analyzes the review text.
- The system returns a sentiment score and identifies positive feedback or pros.
- This helps the vendor quickly understand customer opinions and product reception.

Insert Gemini Sentiment Analysis Screenshot Here

Rule-Based Recommendations & Historical Validation

- Product sales are calculated strictly from genuine transaction records.
- Products are ranked based on the quantity sold.
- The highest-selling products are automatically displayed as recommendations.
- Historical/test transaction data was used to verify that the analytics produce expected, accurate results.

Insert Recommendation / Historical Validation Screenshot Here

ARIMA Demand Forecasting

- Historical sales data is used to identify demand patterns.
- The vendor selects a product.
- An ARIMA model generates a 7-day demand forecast.
- The forecast is compared directly with current inventory levels.
- A restock alert is shown when expected demand is higher than available stock.

Insert ARIMA Forecast Screenshot Here

Milestone 2 Outcome

Customer purchases now generate real transaction data, which feeds the analytics layer. This data is used for customer segmentation, product recommendations, demand forecasting, and review sentiment analysis.

- Real customer ordering and transaction tracking
- Data-driven customer segmentation and recommendations
- 7-day inventory demand forecasting with ARIMA
- Gemini-powered review sentiment analysis

Insert Final Analytics Dashboard Screenshot Here